

Structure and Organization of Public Health Laboratories

1. GENERAL

SCHOOL	PUBLIC HEALTH		
SECTION	PUBLIC HEALTH POLICIES		
LEVEL OF STUDIES	MASTER'S DEGREE IN PUBLIC HEALTH		
COURSE CODE	MSPYL201	SEMESTER OF STUDY	B
COURSE TITLE	STRUCTURE AND ORGANIZATION OF PUBLIC HEALTH LABORATORIES		
INDEPENDENT TEACHING ACTIVITIES <i>in case the credits are awarded in discrete parts of the course e.g. Lectures, Laboratory Exercises, etc. If credits are awarded uniformly for the whole course enter the weekly teaching hours and the total credits</i>		WEEKLY HOURS OF TEACHINGSKALIAS	CREDIT UNITS
Lectures		1	
Laboratory Exercise/ Tutorial Exercise		2	
Add rows if needed. The organisation of teaching and the teaching methods used are described in detail in 4.		3	6
COURSE TYPE <i>Background, General Knowledge, Scientific Area, Skills Development</i>	Scientific area, Skills Development		
PREREQUISITE COURSES:			
LANGUAGE OF TEACHING AND EXAMS:	Greek		
THE COURSE IS OFFERED TO ERASMUS STUDENTS			
ONLINE COURSE (URL) (if available)	www.moodle.uniwa.gr		

2. LEARNING OUTCOMES

Learning Outcomes <i>The learning outcomes of the course are described, the specific knowledge, skills and abilities of an appropriate level that the students will acquire after the successful completion of the course. Consult Annex A</i> • Description of the Level of Learning Outcomes for each cycle of study according to the Qualifications Framework of the European Higher Education Area • Descriptive Indicators of Levels 6, 7 & 8 of the European Qualifications Framework for Lifelong Learning and Annex B • Concise Guide to Writing Learning Outcomes	
After the successful completion of the course the student will be able to: • understand the contribution of the Public Health Laboratory. • clarifies the concepts of diagnosis and surveillance. • analyze the risk of biological parameters. • manage the results of surveillance for public health decision-making. • adapt laboratory capabilities to solving specific problems. • decide on the success of epidemic management with data from laboratory monitoring.	
General Competencies <i>Taking into account the general competencies that the graduate must have acquired (as these are listed in the Diploma Supplement and listed below) which of them is the aim of the course?</i>	
Search, analyze and synthesize data and information, using the necessary technologies Adapting to new situations Decision-making Independent work Teamwork Working in an international environment Working in an interdisciplinary environment Generating new research ideas	Project planning and management Respect for diversity and multiculturalism Respect for the natural environment Demonstration of social, professional and ethical responsibility and gender sensitivity Criticism and self-criticism Promotion of free, creative and inductive thinking

- Search, analysis and synthesis of data and information on the structure, organization and operation of Public Health Laboratories using modern technologies to draw safe conclusions
- Assessment of biological hazards in public health laboratories and leading to decision-making on biosecurity
- Implementation of international quality and accreditation standards
- Conducting autonomous and teamwork on epidemic response issues
- Work in an international and interdisciplinary environment
- Contribution to the production of new research ideas
- Adaptation to new situations and the dynamic and ever-changing scientific landscape to address contemporary challenges

3. COURSE CONTENT

Modules:

1. Introduction to the structure, organization and operation of a Public Health Laboratory
2. Risk assessment and management
3. Biosecurity levels in public health laboratories
4. Implementation of appropriate sampling strategies, packaging and dispatch of biological samples
5. Quality management and international accreditation standards
6. Diagnosis with conventional and molecular techniques
7. Surveillance of infectious diseases - from theory to practice I (Contribution of the Reference Centers)
8. Surveillance of infectious diseases - from theory to practice II (Contribution of the Reference Centers)
9. Surveillance of infectious diseases - from theory to practice III (Contribution of the Reference Centers)
10. The role of the Central Public Health Laboratory
11. Analysis of scenarios for dealing with epidemics
12. Communication of public health laboratories with decision-making centers
13. Enhancing the capabilities of Public Health Laboratories to address contemporary challenges

4. TEACHING AND LEARNING METHODS - EVALUATION

DELIVERY METHOD <i>Face to face, Distance learning, etc.</i>	Group recommendations, laboratory and tutorial exercises								
USE OF INFORMATION AND COMMUNICATION TECHNOLOGIES <i>Use of ICT in Teaching, Laboratory Education, Communication with students</i>	– Posting of course material (notes, lecture slides, exercises, case reports, laboratory quizzes) on the e-learning platform (-). – Use of e-mail and announcements on the e-learning platform for communication with students.								
TEACHING ORGANIZATION <i>The way and methods of teaching are described in detail. Lectures, Seminars, Laboratory Exercise, Field Exercise, Study & analysis of bibliography, Tutorial, Practical (Placement), Clinical Practice, Art Laboratory, Interactive teaching, Educational visits, Study preparation (project), Writing of papers / assignments, Artistic creation, etc.</i>	<table border="1"> <thead> <tr> <th><i>Activity</i></th><th><i>Semester Workload</i></th></tr> </thead> <tbody> <tr> <td>Lectures</td><td>39</td></tr> <tr> <td>Laboratory/Tutorial Exercise</td><td>39</td></tr> <tr> <td>Independent Study</td><td>72</td></tr> </tbody> </table>	<i>Activity</i>	<i>Semester Workload</i>	Lectures	39	Laboratory/Tutorial Exercise	39	Independent Study	72
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Lectures	39								
Laboratory/Tutorial Exercise	39								
Independent Study	72								

<i>The student's study hours for each learning activity are listed as well as the hours of non guided study so that the total workload at semester level corresponds to ECTS standards</i>	<table border="1"> <tr> <td data-bbox="643 206 1029 342">Course Total</td><td data-bbox="1029 206 1302 342">150</td></tr> </table>	Course Total	150
Course Total	150		
STUDENT EVALUATION <i>Description of the evaluation process</i> <i>Assessment Language, Assessment Methods, Formative or Summative, Multiple Choice Test, Short Answer Questions, Essay Development Questions, Problem Solving, Written Assignment, Report/ Report, Oral Examination, Public Presentation, Laboratory Thesis, Clinical Examination of the Patient, Artistic Interpretation, Other / Other</i> <i>Explicitly defined evaluation criteria are mentioned and if and where they are accessible to students.</i>	I. Written final examination (100%) which may include: – Short answer questions – Problem solving – Multiple-choice questions For successful completion, a grade of at least 5/10 is required.		

5. COMPONENT-BIBLIOGRAPHY

<i>- Suggested Bibliography:</i> – WHO strengthening health security by implementing the International Health regulations. Available at: https://www.who.int/ihr/biosafety/en/ – Assoc of Public Health Laboratories. Biorisk Management for Clinical and Public Health Laboratories. Available at: https://www.aphl.org/programs/preparedness/Biosafety-andBiosecurity/Documents/APHL_Biorisk_management_program_guidance_document.pdf – Biorisk management for laboratories and other related organisations. Available at: https://www.iso.org/obp/ui/#iso:std:iso:35001:ed-1:v1:en – WHO Strengthening health security by implementing the International Health Regulations. Guidance on regulations for the Transport of Infectious Substances Guidance on regulations for the transport of infectious substances 2019–2020. available at: https://www.who.int/ihr/publications/WHO-WHE-CPI-2019.20/en/ – Laboratory biosafety manual - World Health Organization available at: https://www.who.int/csr/resources/publications/biosafety/WHO_CDS_CSR_LYO_2004_11/en/ – Biosafety in Microbiological and Biomedical Laboratories, 5th edition, Centers for Disease Control and Prevention and National Institutes of Health, February 2007. http://www.cdc.gov/biosafety/ – Biennial Review of the Lists of Select Agents and Toxins, National Select Agent Registry, CDC. Atlanta, GA. 2010. http://www.selectagents.gov/ – Tzortzopoulos A. Application of principles and tools of Total Quality Management in the health sector – guarantee in the qualitative upgrading of hospitals and the National Health System 2018. Available at: https://www.mednet.gr/archives/2018-5/pdf/649.pdf – World Health Organization Universal access to malaria diagnostic testing: an operational manual – C. Mirande & I. Bizine & A. Giannetti & N. Picot & A. van Belkum Epidemiological aspects of healthcare-associated infections and microbial genomics. European Journal of Clinical Microbiology & Infectious Diseases https://doi.org/10.1007/s10096-017-3170-x – Martin C. J. Maiden, Jane A. Bygraves, Edward Feil, Giovanna Morelli, Joanne E. Russell, Rachel Urwin, Qing Zhang, Jiaji Zhou, Kerstin Zurth, Dominique A. Caugant, Ian M. Feavers, Mark Achtman, and Brian G. Spratt Multilocus sequence typing: A portable approach to the

identification of clones within populations of pathogenic microorganisms. PNAS March 17, 1998 95 (6) 3140-3145; <https://doi.org/10.1073/pnas.95.6.3140>

- Nisarg Gohil, Happy Panchasara, Shreya Patel, Vijai Singh. Molecular Biology Techniques for the Identification and Genotyping of Microorganisms. Microbial Genomics in Sustainable Agroecosystems 2019.
- Kanny Diallo, Martin C. J. Maiden. An Introductory Narrative to the Population Genomics of Pathogenic Bacteria, Exemplified by *Neisseria meningitidis* Microbial Genomics in Sustainable Agroecosystems 2018